

## Supplementary Material

# Toward a Cross-Lingual Romanization Ecosystem for Sinitic Languages: A Paired Mandarin–Cantonese Case Study

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## S1. PhonEngine Interfaces

Figure S1 shows the two PhonSymbol editing modes described in Section 4.2 of the main paper. The final-based mode stores initial, final, syllabic-nasal, and tone mappings, whereas the medial–nucleus–coda mode stores initial, medial, nucleus, coda, syllabic-nasal, and tone mappings.

The figure displays two side-by-side screenshots of the PhonSymbol editing interface. Both interfaces are titled 'UNNAMED' and feature a top navigation bar with buttons for 'Import', 'Export', and 'JSON'.

**Left: final-based mode**

- Initial (sheng\_mu)**: A form with a 'symbol' field and a 'tone' field, with a '+ Add new' button.
- Final (yan\_mu)**: A form with a 'symbol' field and a 'tone' field, with a '+ Add new' button.
- 成音节鼻音 / Syllabic Nasal (sheng\_yin\_mu\_yin)**: A form with a 'symbol' field and a 'tone' field, with a '+ Add new' button.
- 声调 / Tone (sheng\_diao)**: A form with a 'symbol' field and a 'tone' field, with a '+ Add new' button.

**Right: medial–nucleus–coda mode**

- Initial (sheng\_mu)**: A form with a 'symbol' field and a 'tone' field, with a '+ Add new' button.
- Medial (yan\_mu)**: A form with a 'symbol' field and a 'tone' field, with a '+ Add new' button.
- Nucleus (yu\_mu)**: A form with a 'symbol' field and a 'tone' field, with a '+ Add new' button.
- Coda (yan\_mu)**: A form with a 'symbol' field and a 'tone' field, with a '+ Add new' button.
- 成音节鼻音 / Syllabic Nasal (sheng\_yin\_mu\_yin)**: A form with a 'symbol' field and a 'tone' field, with a '+ Add new' button.
- 声调 / Tone (sheng\_diao)**: A form with a 'symbol' field and a 'tone' field, with a '+ Add new' button.

Both interfaces include a 'JSON' button at the bottom right and a 'Preview' button at the bottom left.

Figure S1: *PhonSymbol* interfaces. Left: final-based mode. Right: medial–nucleus–coda mode.

## S2. Using PhonConvert with Kaom.net as a Data Source

The following workflow illustrates how a Kaom.net pronunciation page can be converted into structured romanization data with PhonConvert.

**Step 1.** Open the Kaom.net Sinitic-language pronunciation index at [http://www.kaom.net/si\\_x.php](http://www.kaom.net/si_x.php) (Figure S2) and select a target language location.



Figure S2: Kaom.net page listing supported Sinitic-language locations.

**Step 2.** Click the “音系” (phonology) button in Figure S2 for the selected location to open its phonology page (Figure S3). From Figure S3, click “全字表” (full character table) to open the character-level pronunciation table (Figure S4).

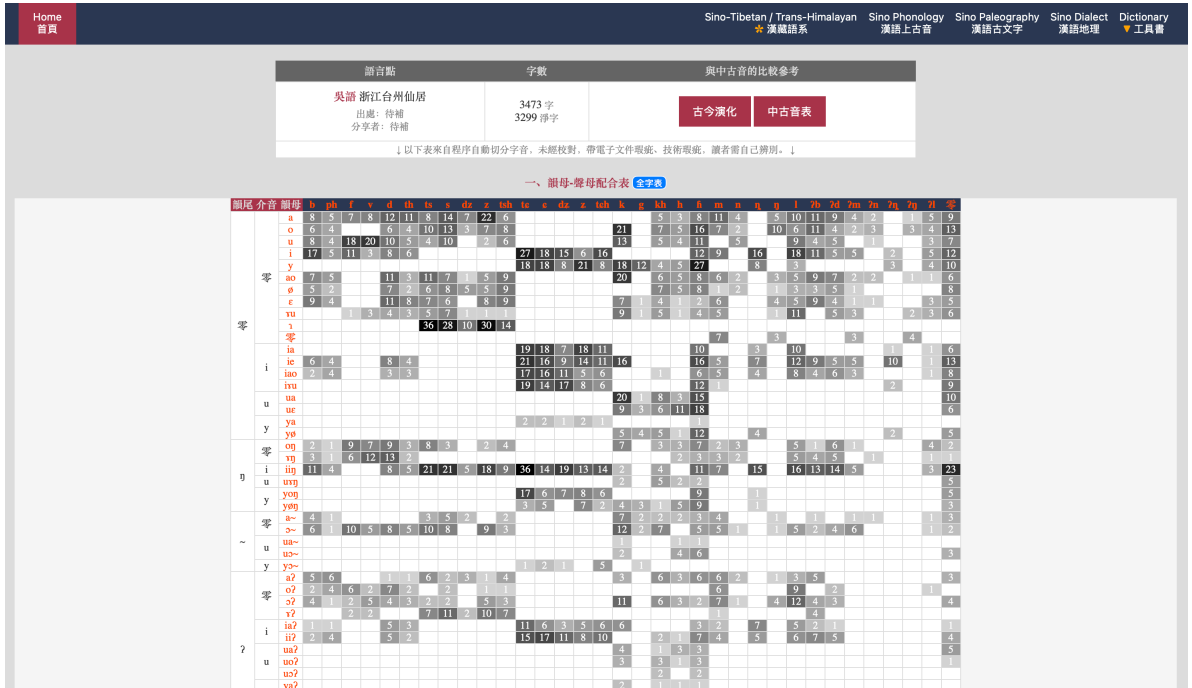


Figure S3: Kaom.net phonology page after selecting a language location.

[illegible]

**Step 3.** On the full character-pronunciation table (Figure S4), select all page content (Ctrl+A on Windows/Linux or Command+A on macOS), copy it, and paste it into the Rich Text Editor on the first page of PhonConvert (Figure S5).

Figure S5: *PhonConvert* input page. *Kaom.net* rich text can be pasted into the Rich Text Editor before data extraction.

PhonConvert phonology-and-romanization organizer after parsing a Kaom.net source page.

Figure S6: *PhonConvert phonology-and-romanization organizer after parsing a Kaom.net source page.*

**Step 5.** PhonConvert can also be used without importing Kaom.net data. From the blank first page of PhonConvert (Figure S5), click “Organizer” to enter a blank phonology-and-romanization organizer (Figure S7) to work from blank or upload a previously saved editing-state JSON file to restore and continue an existing project.

Blank PhonConvert phonology-and-romanization organizer.

Figure S7: *Blank PhonConvert phonology-and-romanization organizer.*



### S3. Additional Phonological Evidence

Table S1: Representative examples of Cantonese /æ:ŋ/ corresponding to /iɔŋ/ in selected neighboring Yue Chinese languages.

| Finals            | 想   | 香   | 將   | 梁   | 釀   |
|-------------------|-----|-----|-----|-----|-----|
| Guangzhou         | æ:ŋ | æ:ŋ | æ:ŋ | æ:ŋ | æ:ŋ |
| Huaxian (Huashan) | iɔŋ | iɔŋ | iɔŋ | iɔŋ | iɔŋ |
| Sanshui (Xi'nan)  | iɔŋ | iɔŋ | iɔŋ | iɔŋ | iɔŋ |
| Xinhui (Huicheng) | iɔŋ | iɔŋ | iɔŋ | iɔŋ | iɔŋ |
| Doumen Zhen       | iɔŋ | iɔŋ | iɔŋ | iɔŋ | iɔŋ |
| Enping (Niujiang) | iɔŋ | iɔŋ | iɔŋ | iɔŋ | iɔŋ |
| Baoan (Shajing)   | iɔŋ | iɔŋ | iɔŋ | iɔŋ | iɔŋ |

**Note.** The table lists representative Chinese characters for which Cantonese /æ:ŋ/ corresponds to /iɔŋ/ in selected neighboring Yue Chinese languages. The data are based on Xiaoxuetang Yue [36]. The table illustrates the historical-phonological motivation for changing Jyutping {oeng} to CantRomZJ1 {eong} and is not intended as a complete comparative survey of Yue reflexes.

### S4. Additional S2R Results and Training Details

Table S2: Cantonese S2R test results under the uncorrected 呢 target condition.

| Seed        | pyjp WER | pyjp CER | romzj1 WER | romzj1 CER |
|-------------|----------|----------|------------|------------|
| 41          | 0.0783   | 0.0498   | 0.0761     | 0.0477     |
| 42          | 0.0742   | 0.0495   | 0.0714     | 0.0446     |
| 43          | 0.0731   | 0.0506   | 0.0704     | 0.0424     |
| <b>Mean</b> | 0.0752   | 0.0500   | 0.0726     | 0.0449     |

Table S3: Training hyperparameters for MMS fine-tuning.

| Setting               | Value               |
|-----------------------|---------------------|
| Model                 | facebook/mms-300m   |
| Objective             | CTC                 |
| Frozen module         | CNN feature encoder |
| Per-device batch size | 8                   |
| Gradient accumulation | 4                   |
| Effective batch size  | 32                  |
| Maximum epochs        | 30                  |
| Learning rate         | 3e-4                |
| Warmup steps          | 1,000               |
| Evaluation interval   | 100 steps           |
| Checkpoint interval   | 100 steps           |
| Checkpoint selection  | Best validation WER |
| Random seeds          | 41, 42, 43          |

Table S4: Cantonese-only S2R results across three random seeds.

| Seed | Jyutping WER | Jyutping CER | CantRomZJ1 WER | CantRomZJ1 CER |
|------|--------------|--------------|----------------|----------------|
| 41   | 0.0805       | 0.0539       | 0.0842         | 0.0591         |
| 42   | 0.0851       | 0.0563       | 0.0833         | 0.0549         |
| 43   | 0.0831       | 0.0533       | 0.0765         | 0.0491         |